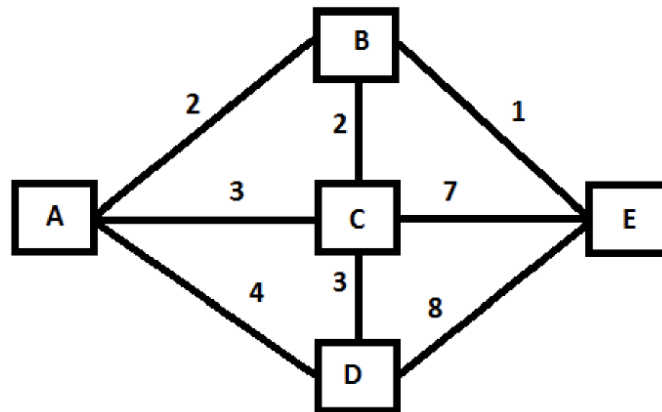


Lab 4 Report

Exercise



1. Step-by-Step Solution

Iteration 1: Initialization

Node E is the destination. The shortest path to itself is 0.

$V(E) = 0$

Iteration 2: Go through node E's neighbors

For Node B

Node B has a direct edge to E with cost 1.

$V(B) = 1$

Path: $E \rightarrow B$

For Node C:

Node C has a direct edge to E with cost 7.

$V(C) = 7$

Path: $E \rightarrow C$

For Node D:

Node D has a direct edge to E with cost 8.

$V(D) = 8$

Path: $E \rightarrow D$

Iteration 3: Go through neighbors to B, C, D

For Node B

B is neighbors to A and C.

A: $2 \text{ (Cost } B \rightarrow A) + 1 \text{ (} V(B)) = 3$

$V(A) = 3$

Path: $E \rightarrow B \rightarrow A$

C: $2 (\text{Cost } B \rightarrow C) + 1 (V(B)) = 3$
 $\text{Min}(3, V(C)) = 3$ (Found a shorter path from C to E)
 $V(C) = 3$
 New path: $E \rightarrow B \rightarrow C$

For Node C

C is neighbors to A, B and D.

A: $3 (\text{Cost } C \rightarrow A) + 3 (V(C)) = 6$
 $\text{Min}(6, 3) = 3$ (Found path is not shorter)

B: Already processed

D: $3 (\text{Cost } C \rightarrow D) + 3 (V(C)) = 6$
 $\text{Min}(6, 8) = 6$ (Found a shorter path from D to E)
 $V(D) = 6$
 New path: $E \rightarrow B \rightarrow C \rightarrow D$

For Node D

D is neighbors to A and C.

A: $4 (\text{Cost } D \rightarrow A) + 6 (V(D)) = 10$
 $\text{Min}(10, V(A) = 3) = 3$ (Found path is not shorter)

C: Already processed

2. Results

To get the shortest path from each node towards E, you reconstruct the path you got before in the other direction. The shortest path values $V(x)$ for all nodes are as follows:

Node	Shortest path to E	$V(x)$
A	$A \rightarrow B \rightarrow E$	3
B	$B \rightarrow E$	1
C	$C \rightarrow B \rightarrow E$	3
D	$D \rightarrow C \rightarrow B \rightarrow E$	6
E	E	0